2026

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SAKE RESTAURANT

Delivery Performance Analysis & Operational Process Redesign

AGENDA

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02 DATA PREPARATION AND SQL ANALYSIS

03 PERFORMANCE ANALYSIS AND KEY FINDINGS

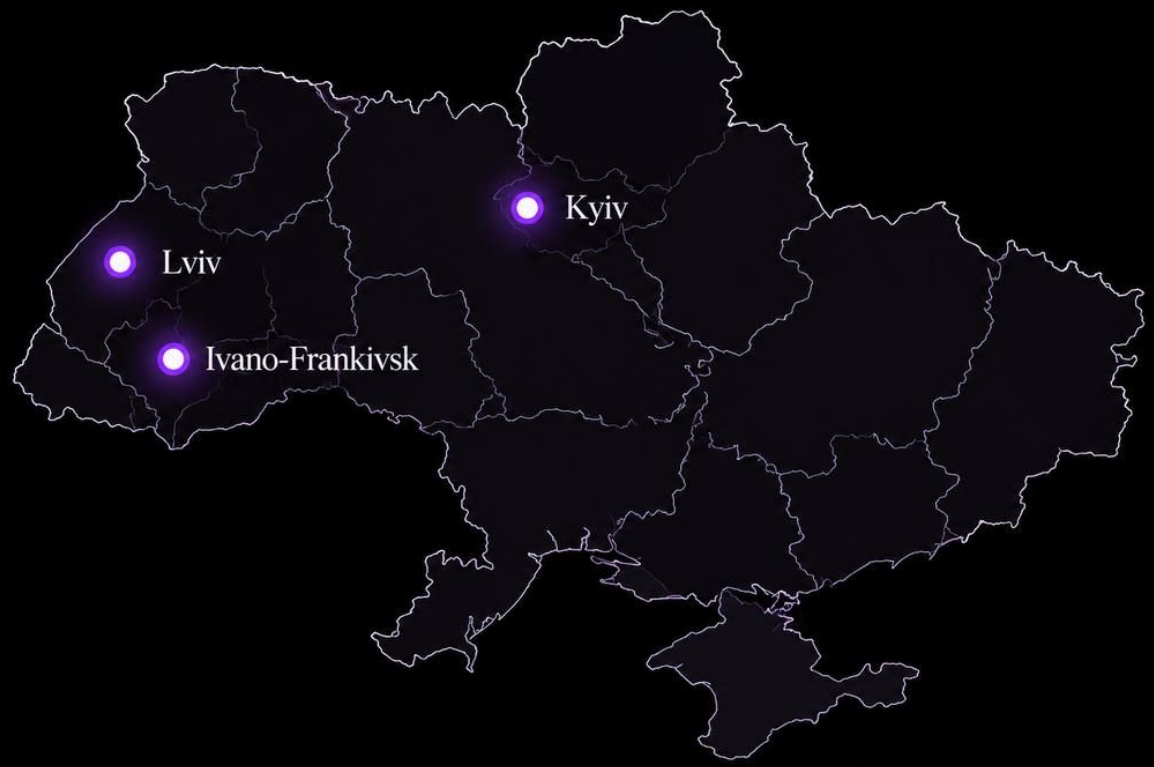
04 BUSINESS PROCESS ANALYSIS

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COMPANY OVERVIEW



SAKE RESTAURANT

SAKE Restaurant is a young Ukrainian fictional restaurant business specializing in Asian cuisine, including sushi, ramen, noodles, rice dishes, and other Asian-inspired meals. The company began as a small restaurant concept and gradually expanded its operations to three Ukrainian cities: Kyiv, Lviv, and Ivano-Frankivsk.

SAKE combines traditional restaurant service with food delivery, allowing customers to enjoy its menu both on-site and through online ordering.

As the business developed, delivery became an important part of its operations, requiring coordination between restaurant staff, couriers, and customer support.

DATA SOURCE AND PREPARATION

DATA SOURCE

For this project, I worked with an order-level operational dataset covering the period from April to July 2026.
The fictional dataset included information about customer orders, delivery times, courier assignments, cancellations, support contacts, and service ratings.

DATA PREPARATION

To prepare the dataset for further analysis, I used Google Sheets. Since this was my first independent Business Analysis project, data preparation was also part of my learning process. While working with the dataset, I practiced using different spreadsheet functions, explored ways to organize and clean data, and gradually became more confident working with larger amounts of information.

This process helped me better understand the structure of the dataset and prepare it for the next stage of the project — SQL analysis and KPI calculations.

DATA SOURCE AND PREPARATION

Order_ID	Order_Date	Month	Restaurant_ID	Courier_ID	Order_Amount_	Order_Status	Preparation_Time_Min	Assignment_Time_Min	Delivery_Time_Min	Customer_Cancelled	Restaurant_Cancelled	Late_15m	Support_Contact	Support_Reason	Courier_Rating	Restaurant_Rating
ORD-00001	2026-04-21	April	R0325	C0429	€34.78	Completed	15	7	39	No	No	No	No		4.7	4.3
ORD-00002	2026-04-16	April	R0078	C0879	€30.44	Customer Cancelled	8	6		Yes	No	No	No		4.9	4.6
ORD-00003	2026-04-18	April	R1284	C1088	€25.09	Completed	16	7	42	No	No	No	No		4.7	4.5
ORD-00004	2026-04-24	April	R0134	C0786	€39.59	Completed	15	8	35	No	No	No	No		4.8	4.3
ORD-00005	2026-04-27	April	R0275		€26.84	Customer Cancelled	24	6		Yes	No	No	Yes	Want to cancel		4.2
ORD-00006	2026-04-11	April	R0269	C0827	€25.61	Completed	20	4	37	No	No	No	No		4.6	4.5
ORD-00007	2026-04-07	April	R1096	C0077	€27.77	Completed	14	1	40	No	No	No	No		4.9	4.2
ORD-00008	2026-04-20	April	R0910	C0041	€36.09	Completed	27	7	32	No	No	No	No		4.4	4.5
ORD-00009	2026-04-13	April	R1066	C0020	€28.33	Completed	17	6	36	No	No	No	No		4.6	4.3
ORD-00010	2026-04-09	April	R0234	C1048	€39.57	Completed	11	2	26	No	No	No	No		4.9	4.4
ORD-00011	2026-04-26	April	R0840	C1072	€22.58	Completed	17	6	38	No	No	No	No		4.9	4.6
ORD-00012	2026-04-10	April	R0183	C0985	€25.58	Completed	18	7	45	No	No	No	No		4.5	4.4
ORD-00013	2026-04-30	April	R0525	C1048	€27.58	Completed	19	7	22	No	No	No	No		4.7	4.7
ORD-00014	2026-04-21	April	R0596	C0205	€33.56	Completed	16	12	35	No	No	No	No		4.9	4.6
ORD-00015	2026-04-12	April	R1326	C0094	€10.10	Completed	16	2	37	No	No	No	No		4.6	4.5
ORD-00016	2026-04-06	April	R0234	C0191	€27.30	Completed	11	9	41	No	No	No	No		4.7	4.3
ORD-00017	2026-04-13	April	R0342	C0980	€32.92	Completed	11	6	41	No	No	No	No		4.4	4.5
ORD-00018	2026-04-04	April	R1166	C0213	€14.61	Completed	19	7	63	No	No	Yes	No		4.7	4.2
ORD-00019	2026-04-22	April	R0843	C0268	€33.06	Completed	10	6	38	No	No	No	No		4.4	4.2
ORD-00020	2026-04-15	April	R1158	C1075	€28.11	Completed	18	5	33	No	No	No	No		5.0	4.4
ORD-00021	2026-04-08	April	R0987	C0196	€15.27	Completed	21	4	33	No	No	No	No		4.9	4.2
ORD-00022	2026-04-12	April	R1036	C0112	€32.94	Completed	12	1	43	No	No	No	No		5.0	4.8
ORD-00023	2026-04-28	April	R0104	C0657	€28.54	Completed	10	5	42	No	No	No	No		5.0	4.4
ORD-00024	2026-04-17	April	R1174	C0482	€50.81	Customer Cancelled	16	8		Yes	No	No	No		4.9	4.5
ORD-00025	2026-04-15	April	R1142	C0037	€36.00	Completed	17	2	52	No	No	No	No		4.9	4.5
ORD-00026	2026-04-18	April	R0935	C0624	€30.93	Completed	15	2	32	No	No	No	No		4.9	4.3
ORD-00027	2026-04-11	April	R0927	C0969	€25.04	Completed	16	5	43	No	No	No	No		4.7	4.3
ORD-00028	2026-04-04	April	R0218	C0598	€37.80	Completed	18	7	37	No	No	No	No		4.5	4.4
ORD-00029	2026-04-02	April	R0385	C0839	€35.36	Completed	12	6	28	No	No	No	No		4.9	4.6
ORD-00030	2026-04-04	April	R1297	C0693	€35.14	Completed	13	1	40	No	No	No	No		4.9	4.5
ORD-00031	2026-04-27	April	R1087	C1033	€24.83	Completed	13	7	48	No	No	Yes	No		4.9	4.6
ORD-00032	2026-04-24	April	R1185	C1042	€47.09	Completed	18	6	31	No	No	No	No		4.8	4.0
ORD-00033	2026-04-06	April	R1248	C0660	€26.42	Completed	13	2	42	No	No	No	No		4.5	4.4
ORD-00034	2026-04-30	April	R0846	C0603	€24.60	Completed	12	3	40	No	No	No	No		4.6	4.4
ORD-00035	2026-04-20	April	R1296	C0643	€22.38	Completed	9	3	46	No	No	No	No		4.6	4.0
ORD-00036	2026-04-07	April	R0365	C0272	€30.82	Completed	20	4	42	No	No	No	No		5.0	4.4
ORD-00037	2026-04-17	April	R0855	C0984	€14.64	Completed	13	3	42	No	No	Yes	No		4.3	4.2
ORD-00038	2026-04-22	April	R0052		€26.77	Customer Cancelled	7	3		Yes	No	No	Yes	Want to cancel		4.8
ORD-00039	2026-04-13	April	R0300	C0330	€35.20	Completed	20	6	46	No	No	No	No		5.0	4.4
ORD-00040	2026-04-27	April	R1151	C0735	€41.58	Completed	18	5	44	No	No	No	No		4.8	4.3
ORD-00041	2026-04-28	April	R0782	C0590	€29.09	Completed	21	2	41	No	No	No	No		4.7	4.4
ORD-00042	2026-04-13	April	R1271	C0111	€13.27	Completed	19	2	22	No	No	No	No		4.9	4.4
ORD-00043	2026-04-28	April	R1236	C0754	€30.51	Completed	7	7	25	No	No	No	No		4.7	4.2
ORD-00044	2026-04-07	April	R1053	C0153	€30.80	Completed	5	4	40	No	No	No	No		4.8	4.5
ORD-00045	2026-04-04	April	R0318	C0523	€20.18	Completed	18	3	36	No	No	No	No		5.0	4.4
ORD-00046	2026-04-02	April	R1005		€31.52	Customer Cancelled	18	10		Yes	No	No	Yes	Where is courier		4.6
ORD-00047	2026-04-26	April	R1035	C0451	€19.39	Completed	13	5	32	No	No	No	No		5.0	4.4
ORD-00048	2026-04-06	April	R0602	C0437	€17.35	Completed	18	6	39	No	No	No	No		4.6	4.3

BUSINESS PERFORMANCE - APRIL-JULY 2026

KPI	April	May	June	July
Total Orders	2323	2436	2578	2663
Completed Orders	2191	2290	2403	2394
Customer Cancellation Rate	3.79%	3.98%	4.69%	7.89%
Restaurant Cancellation Rate	1.89%	2.01%	2.09%	2.22%
Average Order Amount (€)	€26.97	€27.28	€28.36	€28.16
Average Delivery Time (min)	38.33	39.35	42.45	51.76
Late >15 min Rate	8.63%	9.65%	13.94%	28.91%
Support Contacts	119	127	161	278
Support Contact Rate	5.12%	5.21%	6.25%	10.44%
Average Courier Rating	4.71	4.68	4.60	4.38
Average Restaurant Rating	4.42	4.42	4.41	4.40
Average Preparation Time	14.94	15.65	16.13	16.51
Average Assignment Time	4.47	4.85	6.54	10.49

OPERATIONAL CHANGES - JUNE 2026

01

COURIER COMPENSATION REDUCTION

In June, SAKE Restaurant reduced the base compensation paid to couriers per completed order by 8%.
This change introduced a potential workforce-related factor for the analysis, particularly regarding courier availability, order acceptance, and motivation.

02

COURIER ASSIGNMENT ALGORITHM REPLACEMENT

The company completely replaced its previous courier assignment algorithm with a new order allocation system.
The internal logic of the new algorithm was not provided in the initial business data, making its potential influence on assignment efficiency an important subject for further investigation.

03

MARKETING BUDGET INCREASE

In July, SAKE Restaurant increased its marketing budget from €110,000 to €190,000 — an increase of approximately 72.7%.
The change was considered alongside the company's growing order volume when evaluating operational capacity and delivery performance.

DATA ANALYSIS WITH SQL

After preparing the dataset in Google Sheets, I moved on to the next stage of the project — data analysis using SQL.

My initial knowledge of SQL came from computer science classes at my German Gymnasium. However, this project gave me the opportunity to apply those fundamentals to a larger dataset and a practical business problem.

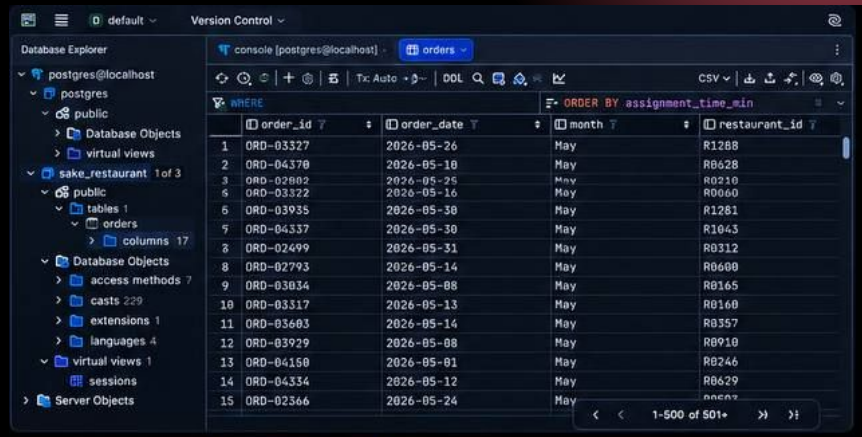
I used SQL to explore the prepared data, calculate key performance indicators (KPIs), and compare operational performance across April, May, June, and July 2026.

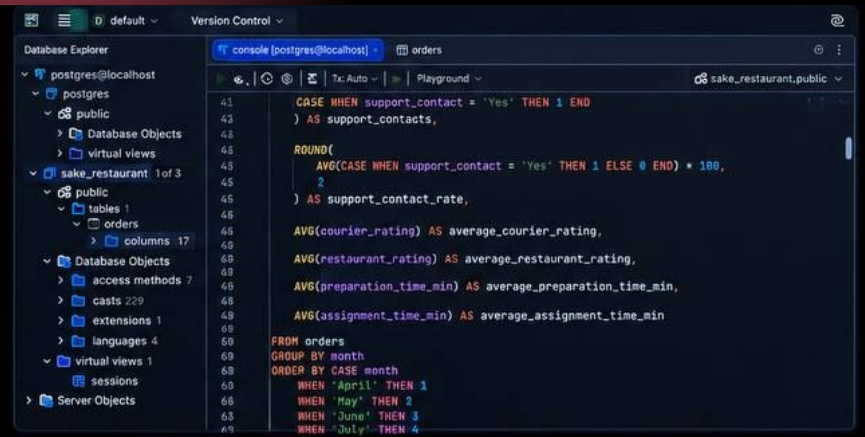
The analysis focused on delivery times, courier assignment efficiency, order cancellations, customer support activity, and service ratings.

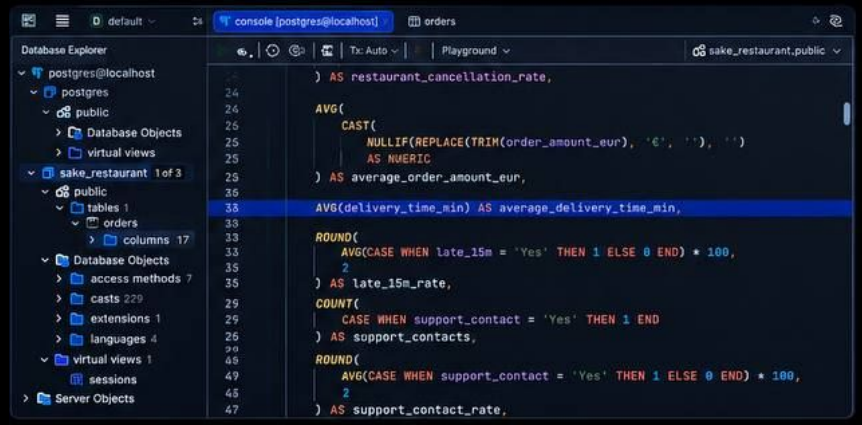
The results formed the basis for identifying operational issues and investigating potential areas for improvement.

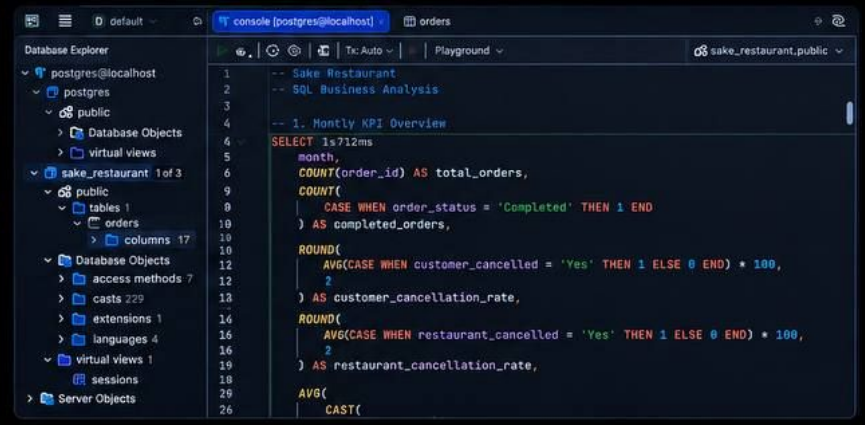
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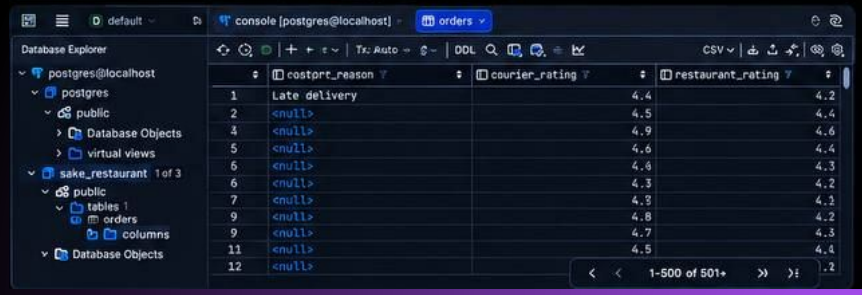
DATA ANALYSIS WITH SQL

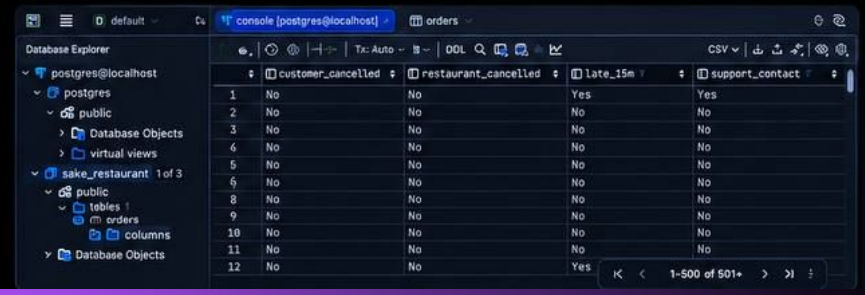












BUSINESS PROBLEM IDENTIFICATION

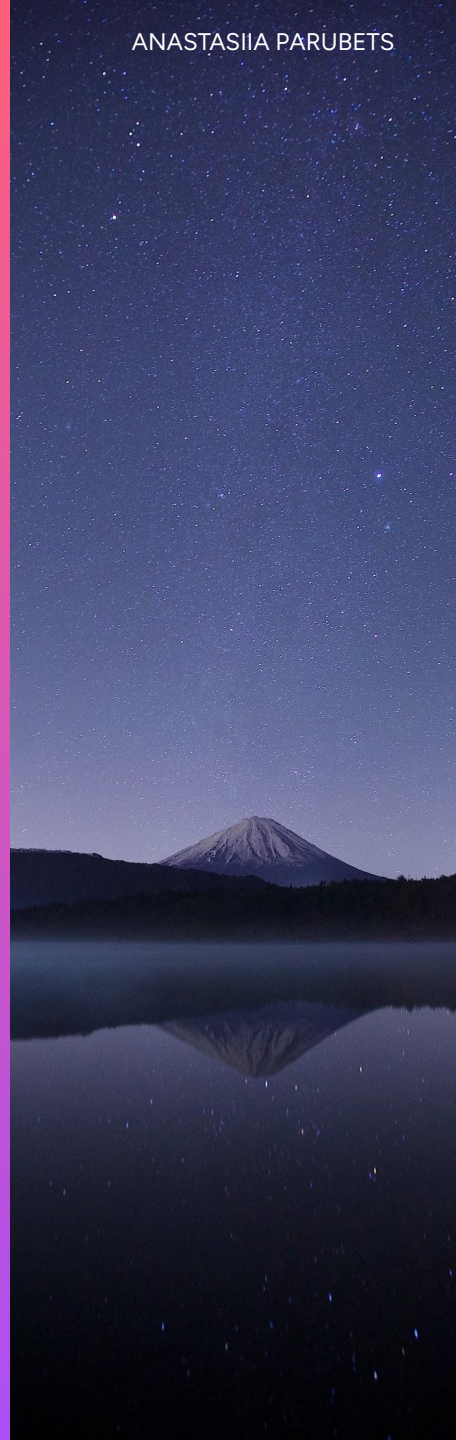
Between April and July 2026, SAKE Restaurant experienced a deterioration in delivery performance despite a continued increase in order volume.

The most significant changes occurred between June and July:

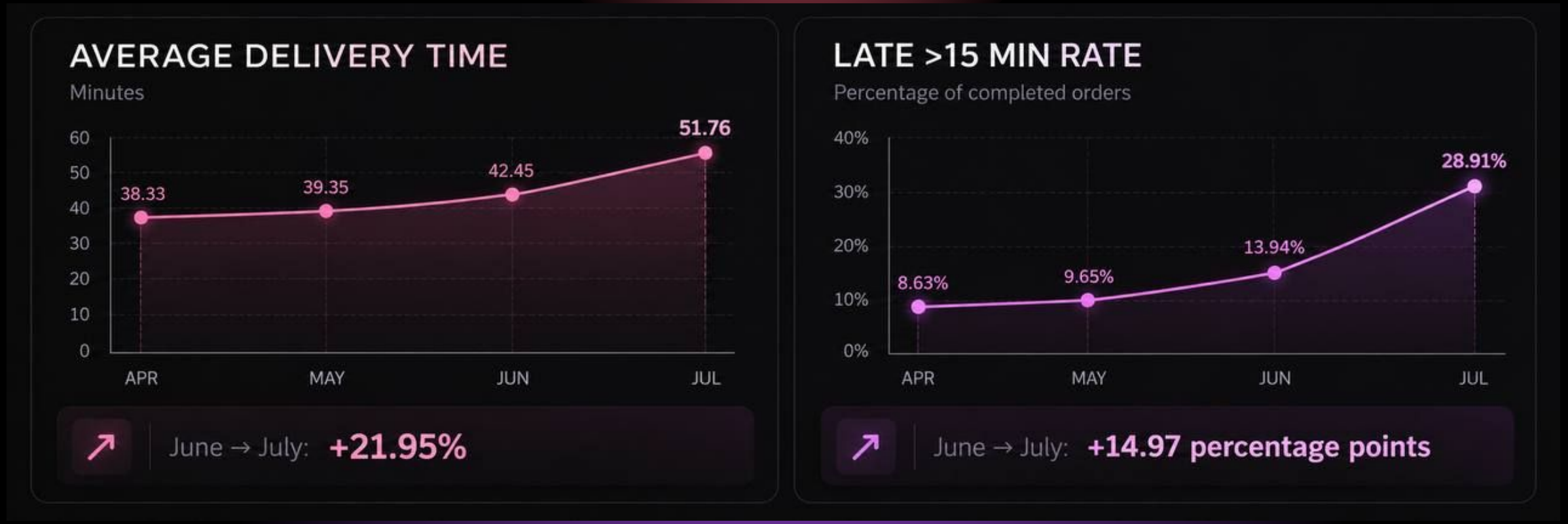
- Average delivery time: 42.45 → 51.76 minutes (+21.95%)
- Deliveries delayed by more than 15 minutes: 13.94% → 28.91% (+14.97 percentage points)
- Customer cancellation rate: 4.69% → 7.89% (+3.20 percentage points)
- Support contact rate: 6.25% → 10.44% (+4.19 percentage points)
- Average courier assignment time: 6.54 → 10.49 minutes (+60.40%)

At the same time, completed orders decreased slightly, even as total order volume continued to grow.

The *main business problem* was to identify which stages of the delivery process contributed most to the deterioration in performance and determine how the process could be improved.



DELIVERY PERFORMANCE DETERIORATION



Average delivery time increased from 42.45 minutes in June to 51.76 minutes in July (+21.95%), indicating a significant decline in delivery efficiency. The share of orders delayed by more than 15 minutes rose from 13.94% to 28.91% (+14.97 percentage points), showing that delivery delays became substantially more frequent.

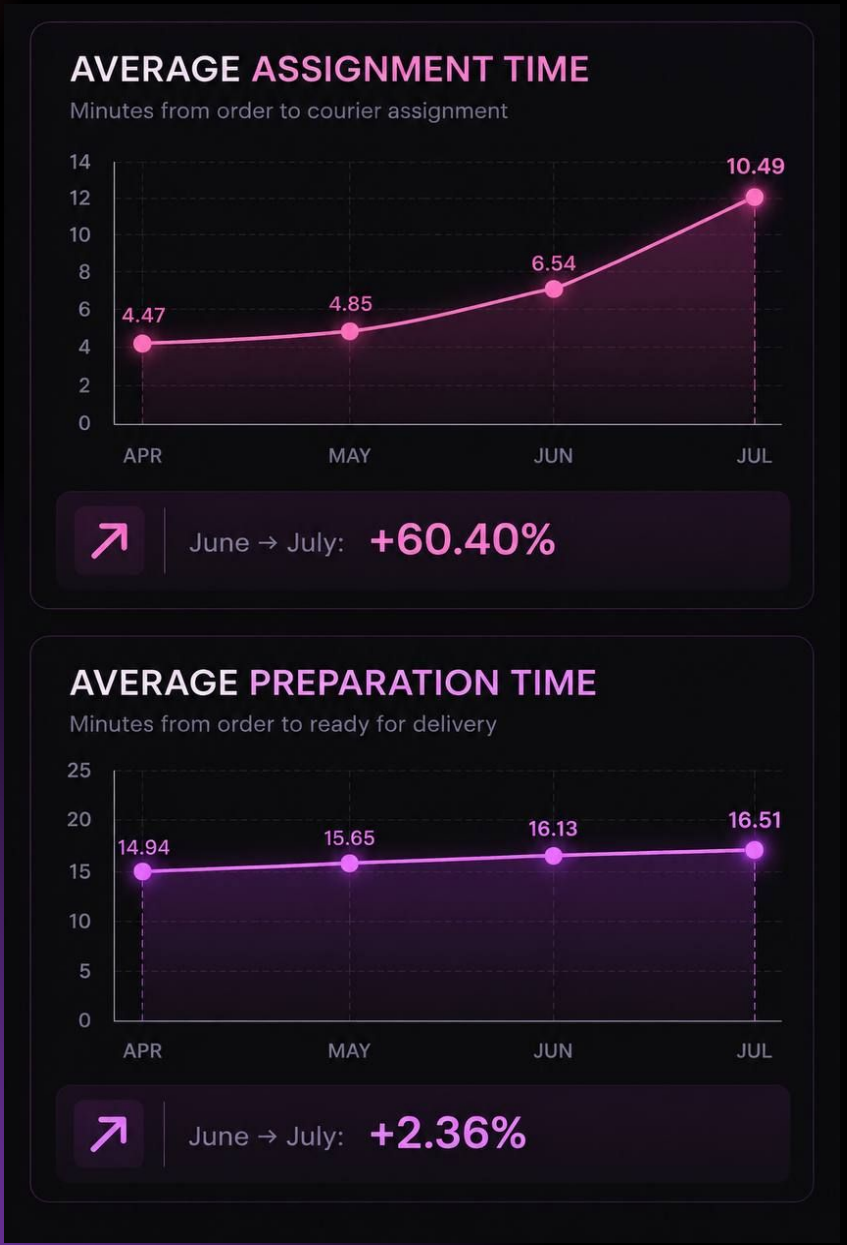
Key Takeaway: Both indicators point to a serious deterioration in delivery performance, particularly between June and July.

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THE BOTTLENECK LOCALIZATION

Average courier assignment time increased from 6.54 minutes in June to 10.49 minutes in July (+60.4%), suggesting growing inefficiency in the courier allocation process. Average food preparation time increased only slightly, from 16.13 to 16.51 minutes (+2.4%), indicating that kitchen operations remained relatively stable.

Key Takeaway : Courier assignment time increased much faster than food preparation time, making the **courier allocation process** a key area for further investigation.

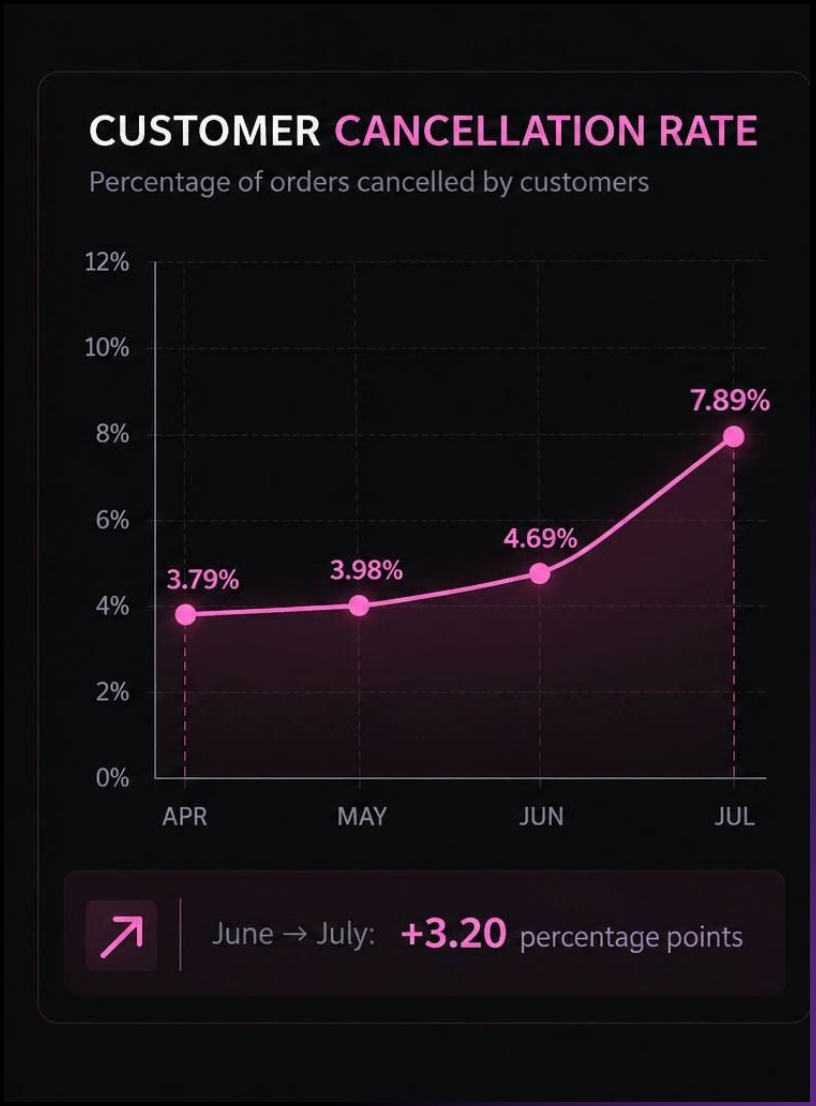


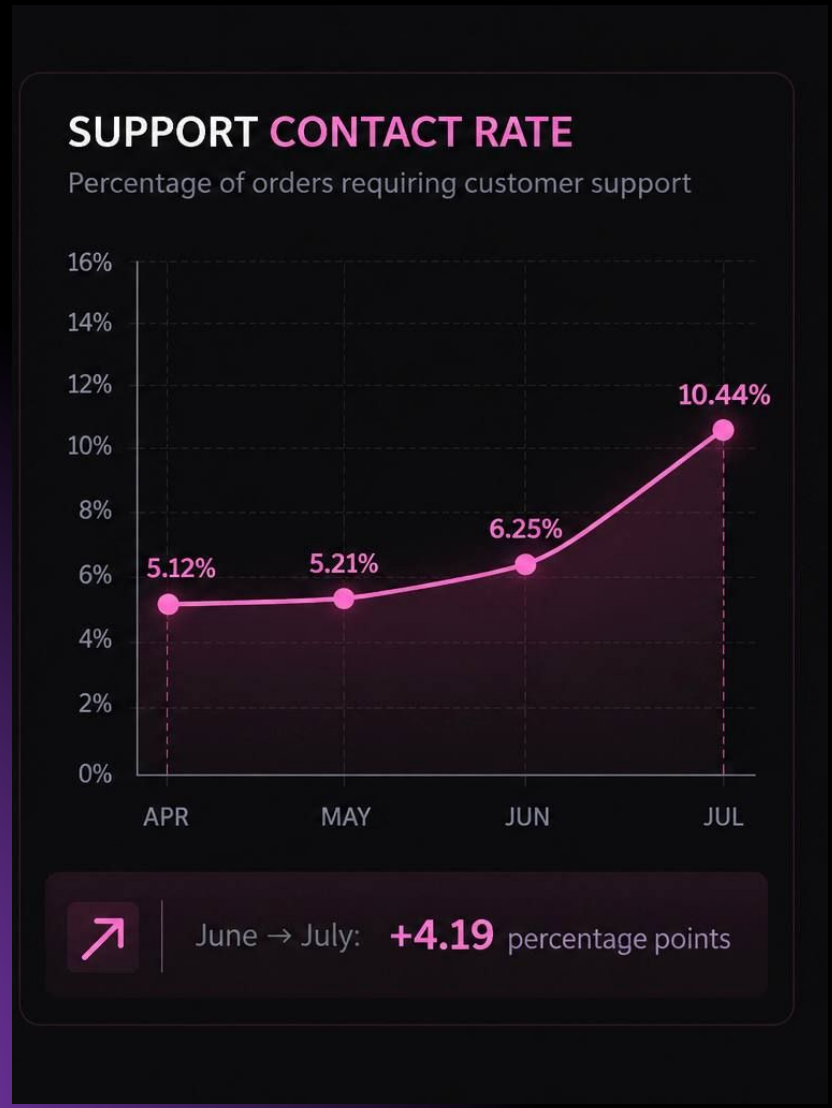
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CUSTOMER IMPACT

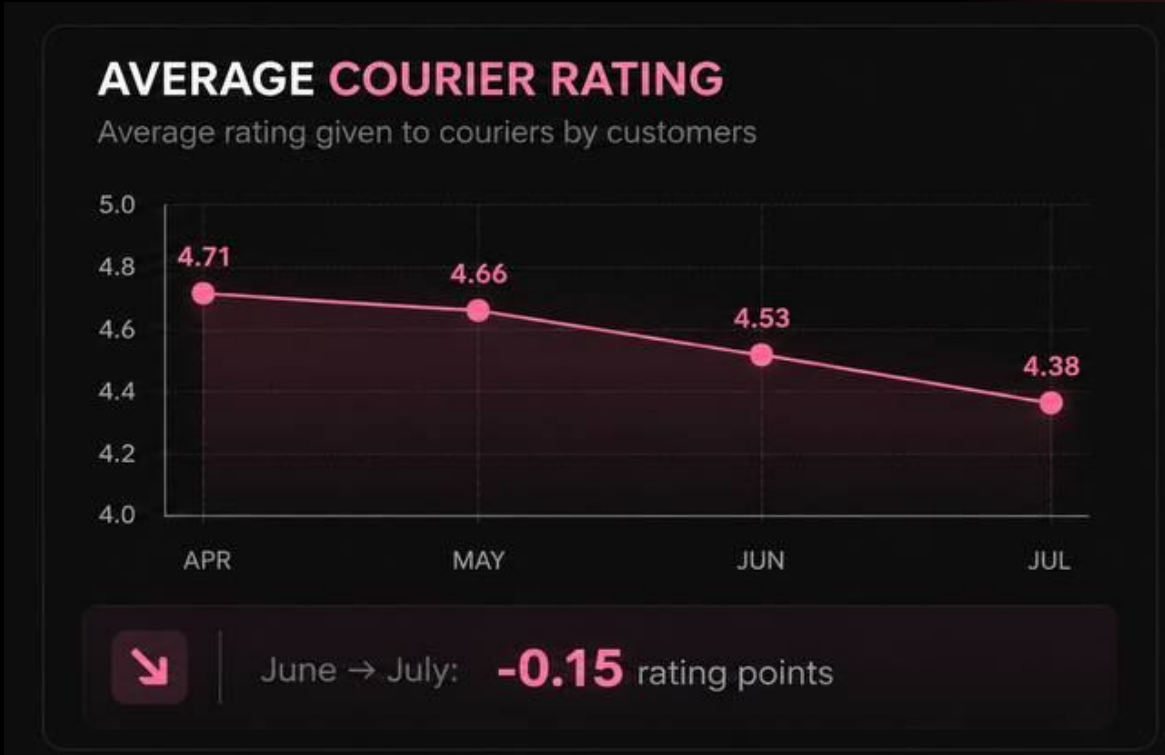
Customer cancellations increased from 4.69% in June to 7.89% in July (+3.20 percentage points), indicating that more customers abandoned their orders. The share of orders requiring customer support increased from 6.25% to 10.44% (+4.19 percentage points), placing additional pressure on the support team.

Key Takeaway : As delivery performance deteriorated, customer cancellations and support requests increased, suggesting a broader impact on customer experience and operational efficiency.





COURIER PERFORMANCE VS. RESTAURANT PERFORMANCE





Average courier rating declined from 4.60 in June to 4.38 in July, indicating a deterioration in customer satisfaction with delivery services. Meanwhile, the average restaurant rating remained relatively stable (4.41 → 4.40), suggesting that the decline in overall service quality was more closely associated with delivery operations than restaurant performance.

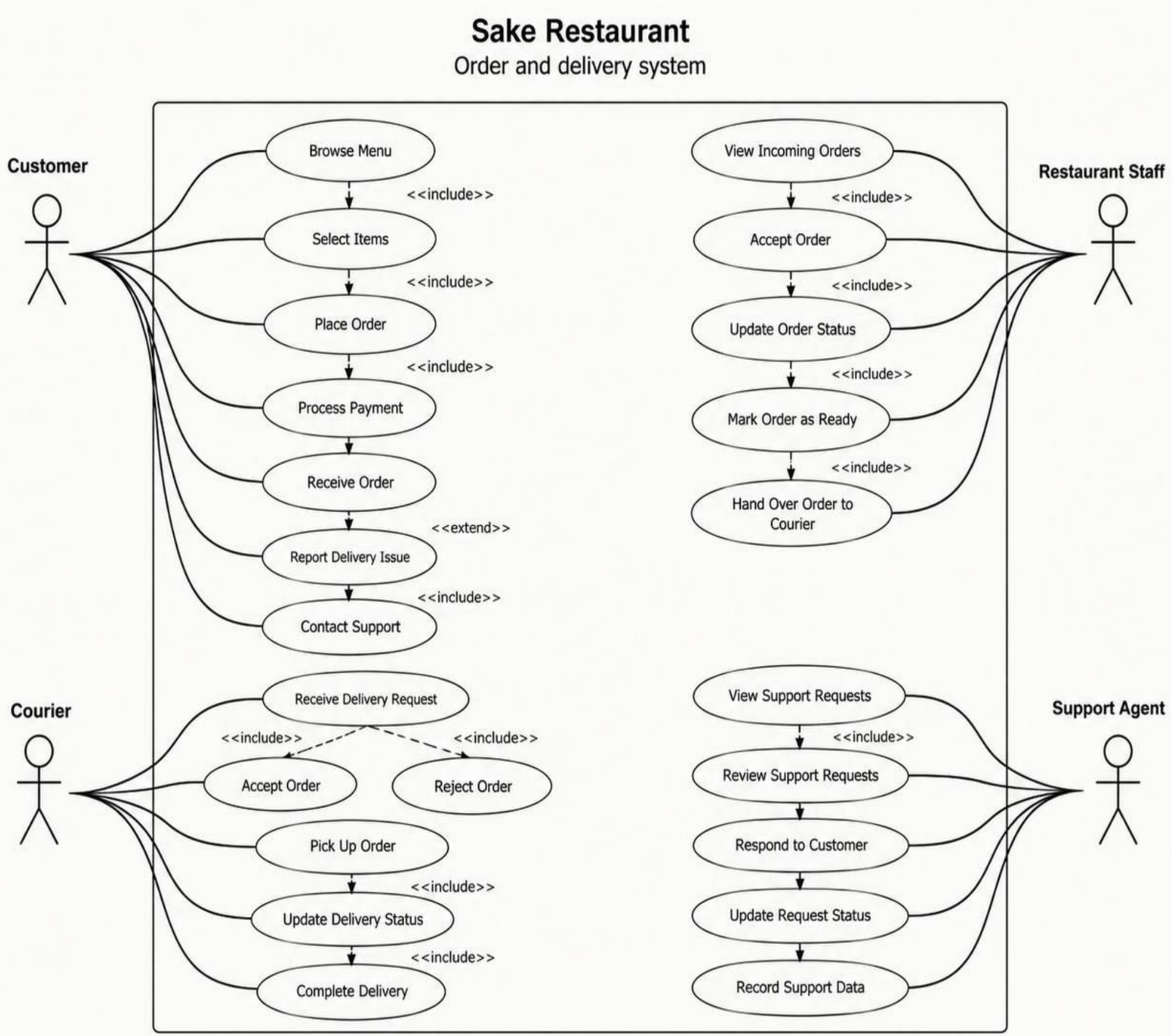
Key Takeaway : **Delivery operations** appear to be the primary area of concern.

CURRENT BUSINESS PROCESS

System Overview - Use Case Diagram

The current SAKE Restaurant system connects customers, restaurant staff, couriers, and customer support throughout the order fulfilment process. Each participant performs specific functions, from placing and preparing orders to coordinating deliveries and resolving customer issues.

To understand where operational inefficiencies may occur, I first mapped the existing system and its key interactions before examining the delivery workflow in greater detail.



CURRENT BUSINESS PROCESS

The Business Idea Behind the New Courier Assignment Algorithm

In June 2026, SAKE Restaurant introduced a **new courier assignment algorithm** as part of its operational changes.

The business idea behind this decision was to give lower-rated couriers more opportunities to complete deliveries and potentially improve their performance ratings.

Instead of prioritizing delivery efficiency, the new assignment logic placed greater emphasis on **courier ratings**, giving preference to couriers with lower scores.

While the initiative was intended to support courier development, it raised an important business question: could prioritizing lower-rated couriers negatively affect delivery performance and customer experience?

CURRENT BUSINESS PROCESS

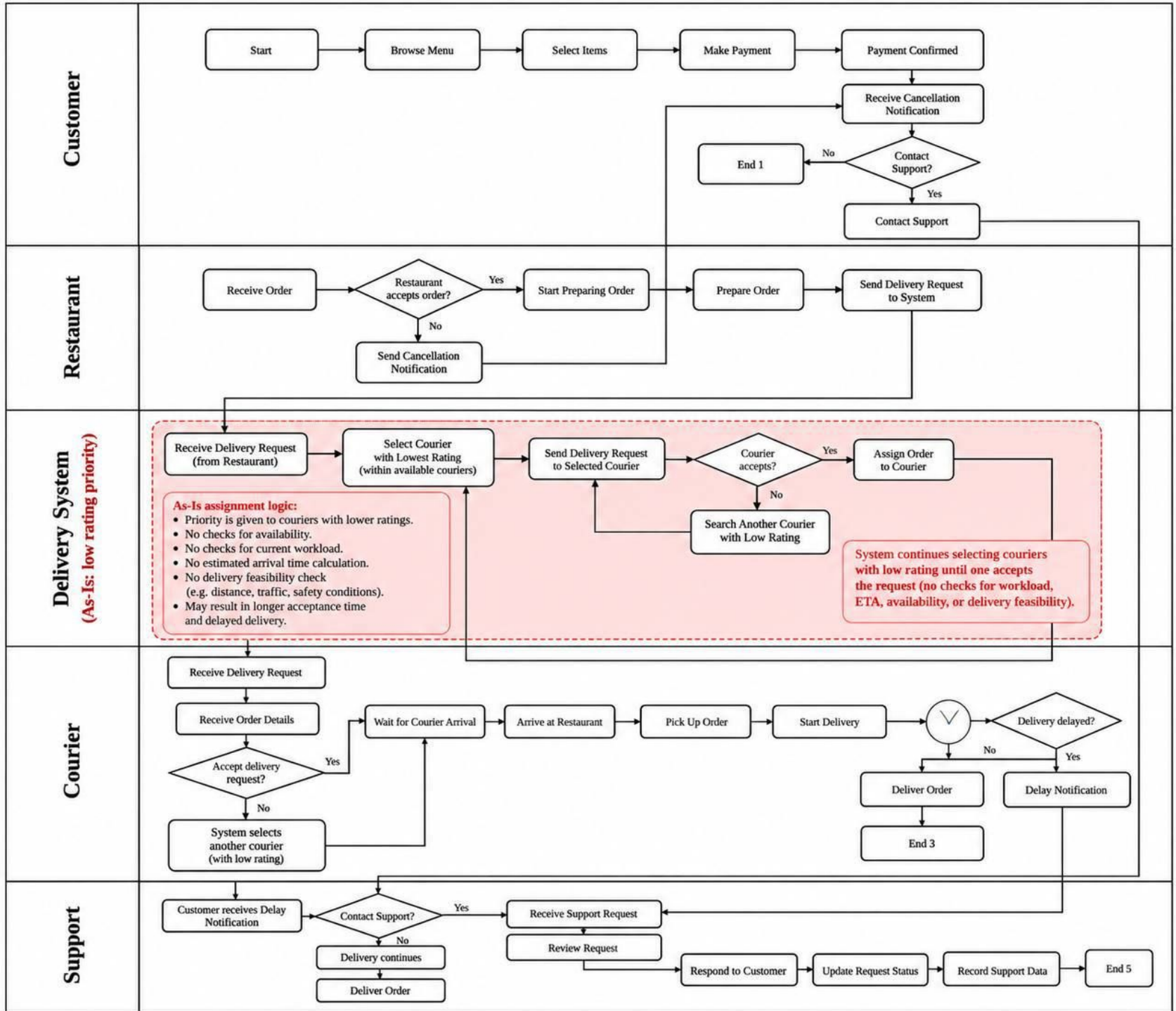
System Overview - AS-IS Diagram

The As-Is process illustrates the assumed logic of the courier assignment system introduced in June 2026.

When a restaurant submits a delivery request, the system prioritizes couriers with lower ratings. If the selected courier rejects the request, the system searches for another lower-rated courier and repeats the process until the order is accepted.

The modeled process does not include explicit checks for courier workload, estimated arrival time, or delivery feasibility.

These potential weaknesses may contribute to longer assignment times and delivery delays.



ROOT CAUSE ANALYSIS

As illustrated in the chart, both average preparation time and average assignment time increased between April and July 2026. However, their growth patterns were significantly different.

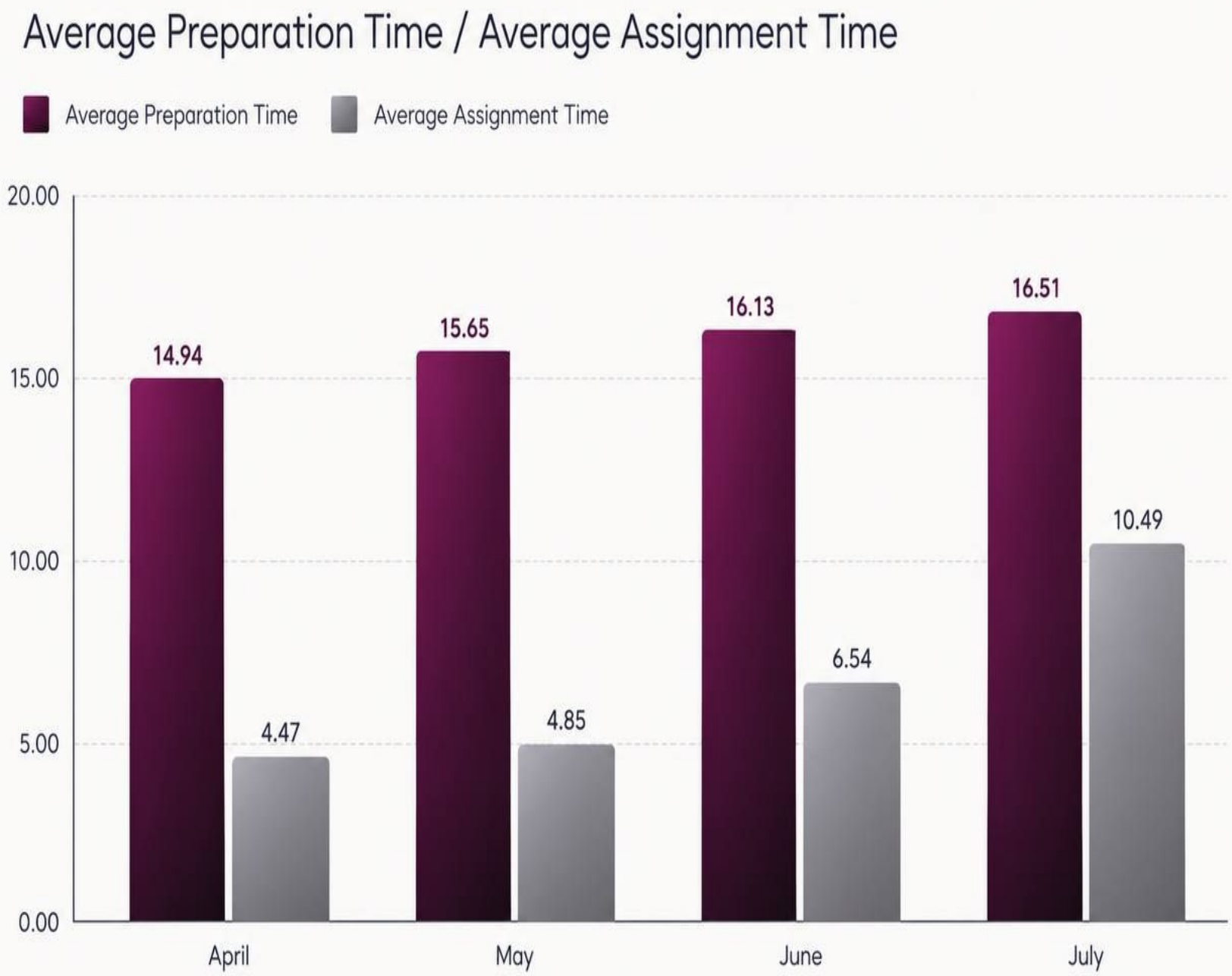
Average preparation time increased gradually throughout the period, from 14.94 minutes in April to 16.51 minutes in July.

In contrast, average assignment time experienced a much sharper increase, rising from 4.47 minutes in April to 10.49 minutes in July.

The most noticeable difference occurred between June and July, when average assignment time increased by 60.4%, while average preparation time increased by only 2.4%.

These findings suggest that the **assignment process** may represent a significant operational bottleneck, as the time required to assign orders increased considerably faster than the time needed to prepare them.

However, further investigation is required to determine whether the new courier assignment algorithm contributed directly to this increase.



KEY FINDINGS

SQL ANALYSIS

The analysis revealed a growing gap between order demand and operational performance. From June to July, total orders increased by 3.30%, while completed orders decreased by 0.37%.

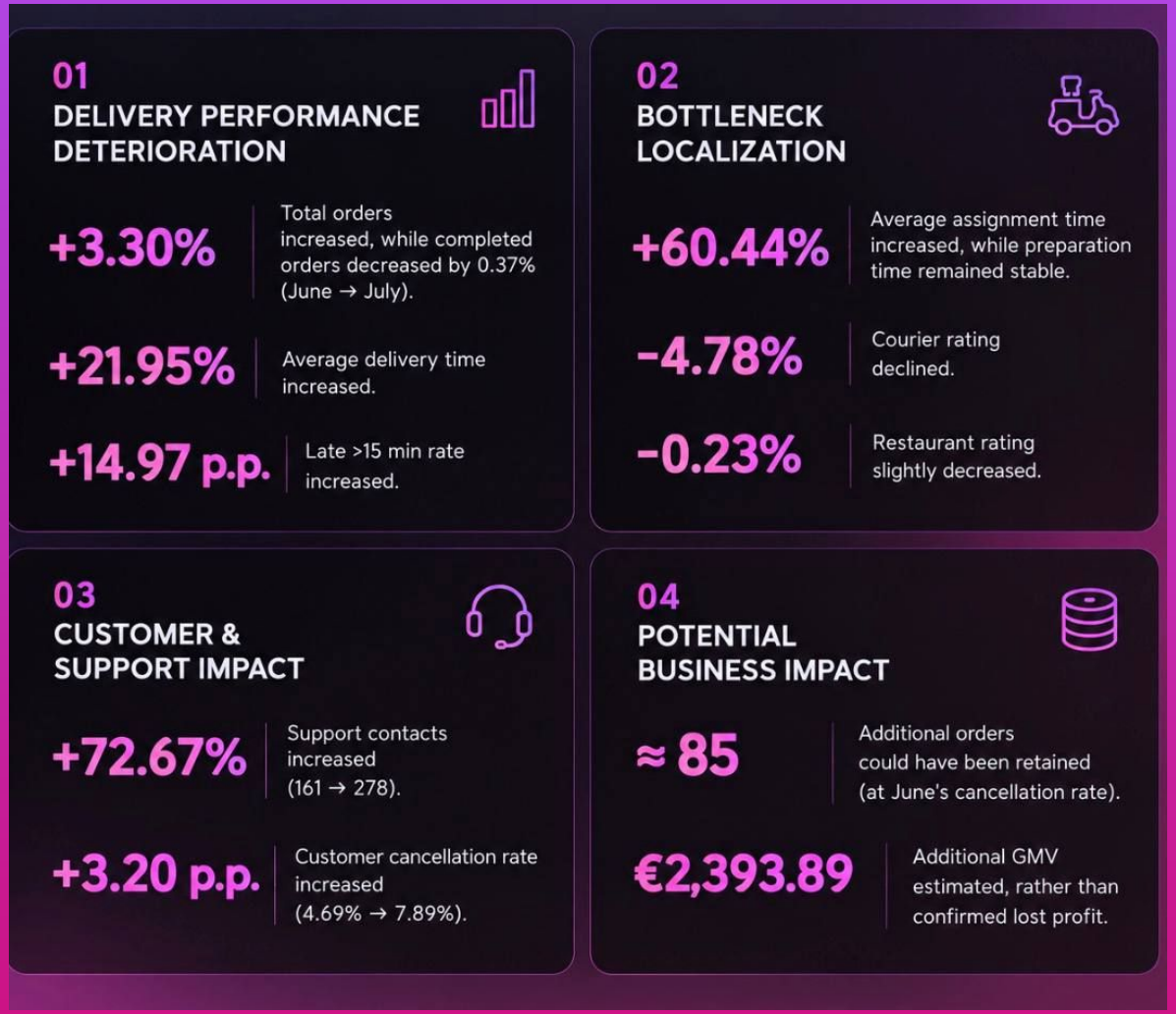
Average delivery time rose by 21.95%, and the share of deliveries delayed by more than 15 minutes increased by 14.97 percentage points.

To investigate this deterioration, I compared restaurant preparation and courier assignment times. While preparation time increased by only 2.36%, assignment time rose by 60.44%, suggesting a potential bottleneck in the courier assignment process.

The impact extended beyond delivery operations. Support contacts increased by 72.67%, while the customer cancellation rate rose by 3.20 percentage points.

Using June's cancellation rate as a reference, I estimated that approximately 85 additional orders could potentially have been retained in July, representing around €2,393.89 in additional gross merchandise value.

Overall, the findings highlight courier assignment as a priority for further investigation and operational improvement.121:54



PROCESS WEAKNESSES

Courier Selection Priority

The new assignment algorithm prioritizes couriers with lower ratings rather than selecting them primarily based on delivery efficiency.

This approach may improve access to orders for lower-rated couriers, but it does not necessarily ensure optimal delivery performance.

Operational Checks

The modelled assignment process does not include explicit checks for courier workload, estimated arrival time or delivery feasibility.

As a result, an order may be offered to a courier who is not the most suitable candidate for that delivery.

Reassignment Process

When a courier rejects a delivery request, the system continues searching for another lower-rated courier.

Repeated assignment attempts may increase the time required to find a courier and delay the beginning of delivery.

Limited Delivery Reliability Controls

The current process does not explicitly prioritize delivery feasibility or expected delivery time during courier selection.

This may increase the risk of late deliveries, customer cancellations and additional support requests.

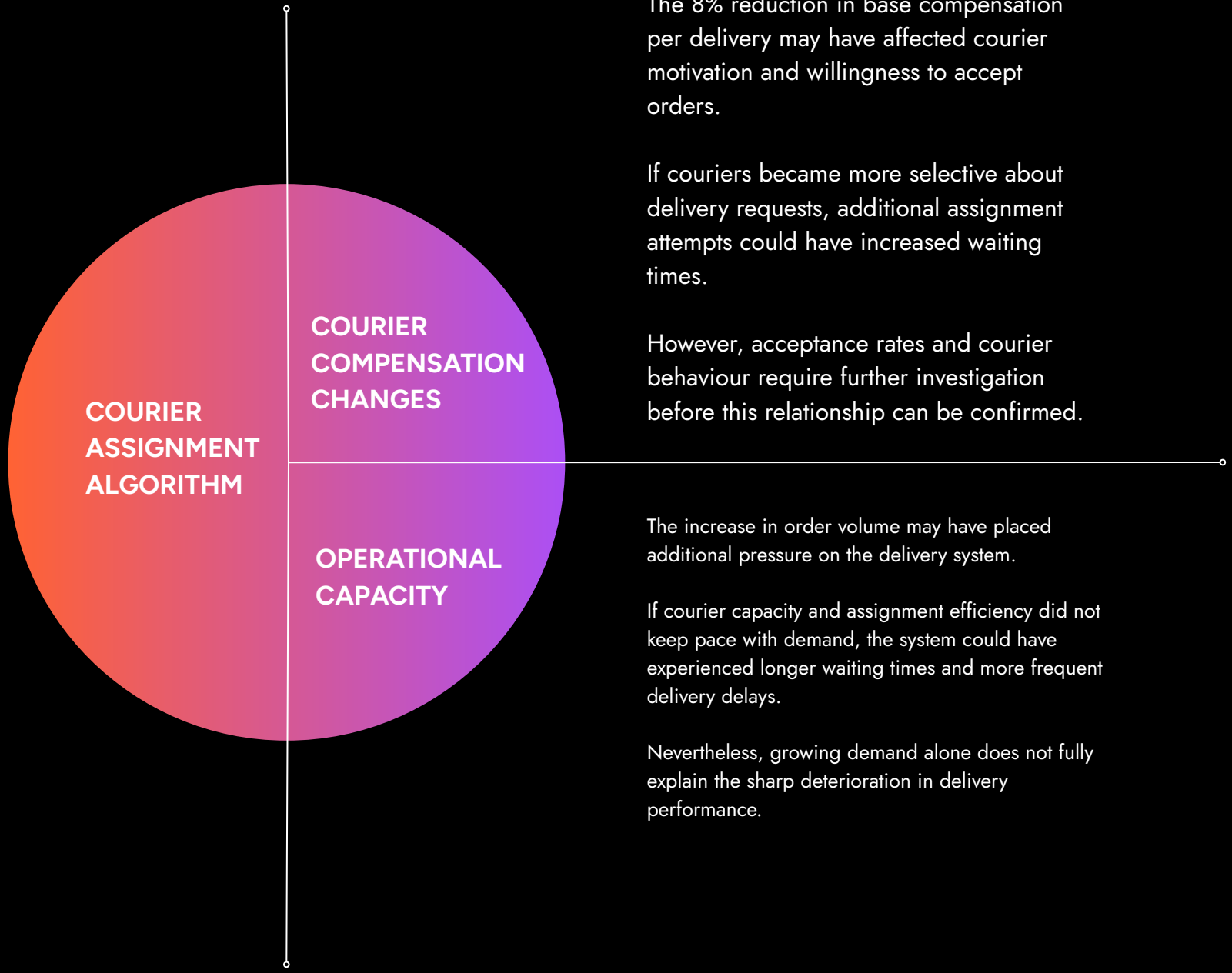
ANASTASIIA PARUBETS

KEY HYPOTHESES

The introduction of a new courier assignment algorithm may have contributed to longer assignment times.

By prioritizing lower-rated couriers without sufficient consideration of operational efficiency, the system may require more time to identify a suitable courier.

This hypothesis is supported by the 60.44% increase in average assignment time between June and July.



The 8% reduction in base compensation per delivery may have affected courier motivation and willingness to accept orders.

If couriers became more selective about delivery requests, additional assignment attempts could have increased waiting times.

However, acceptance rates and courier behaviour require further investigation before this relationship can be confirmed.

The increase in order volume may have placed additional pressure on the delivery system.

If courier capacity and assignment efficiency did not keep pace with demand, the system could have experienced longer waiting times and more frequent delivery delays.

Nevertheless, growing demand alone does not fully explain the sharp deterioration in delivery performance.

PROPOSED SOLUTIONS

01 Introduce Eligibility Checks

Before sending a delivery request, the system should evaluate courier availability, current workload, estimated arrival time and delivery feasibility. Courier ratings may remain a secondary consideration, but they should not override operational requirements.

02 Implement a Two-Stage Search

Stage 1 — Local Search: The system searches for suitable couriers within the local delivery zone for up to 3 minutes and 45 seconds. Stage 2 — Expanded Search: If no suitable courier is found within this period, the system expands the search to all eligible couriers in the delivery area.

03 Improve Reassignment Efficiency

If a courier rejects a delivery request, the system moves to the next eligible candidate while maintaining the same operational checks. This approach is intended to reduce unnecessary assignment attempts and improve delivery reliability.

04 Maintain Customer Transparency

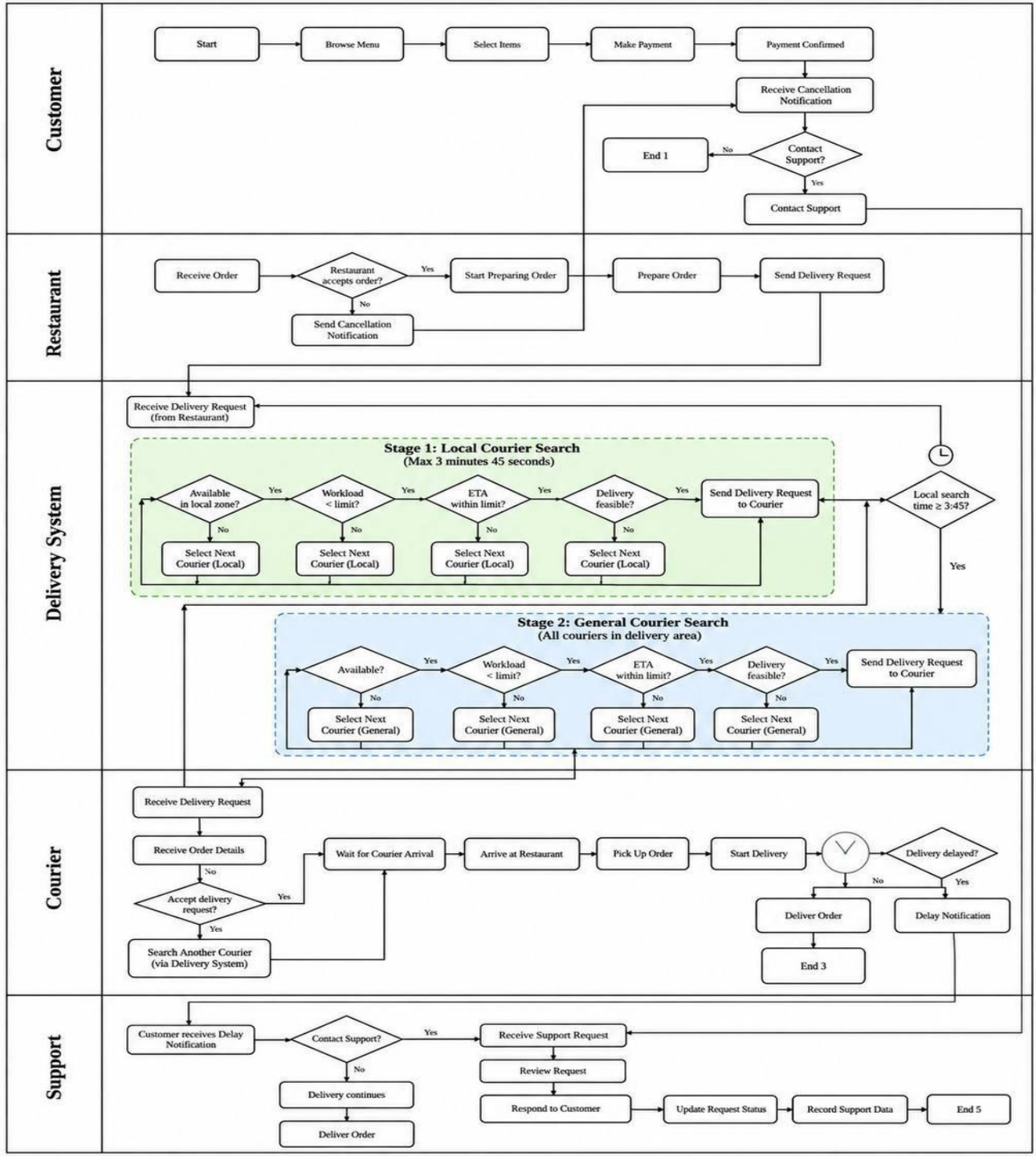
Customers should receive clear order-status updates while the system searches for a courier. The proposed solution retains a fixed delivery price without introducing courier bidding.

TO-BE Business Process

The proposed To-Be process introduces operational eligibility checks and a two-stage courier search to improve assignment efficiency.

Unlike the modelled As-Is process, courier selection prioritizes availability, workload, estimated arrival time and delivery feasibility.

The redesigned process aims to reduce assignment delays while preserving fair opportunities for couriers.



PILOT TESTING

PILOT IMPLEMENTATION

The optimized algorithm would initially be introduced in a limited number of delivery zones.

Where possible, comparable zones operating under the existing algorithm would serve as a control group.

PERFORMANCE MONITORING

During the pilot, I would monitor the following key performance indicators:

- Average assignment time
- Average delivery time
- Late delivery rate (>15 minutes)
- Customer cancellation rate
- Support contact rate
- Courier acceptance and rejection rates
- Courier ratings

RESULTS EVALUATION

After the pilot, I would compare the results with baseline performance and, where available, the control group.

The analysis would help determine whether the proposed changes improved delivery efficiency without negatively affecting courier participation or service quality.

PILOT -> MONITOR -> EVALUATE -> SCALE

SAKE EMPLOYEE BENEFITS PROGRAM



SAKE EMPLOYEE BENEFITS PROGRAM

Following the 8% reduction in base courier compensation, I explored an additional initiative aimed at supporting courier satisfaction and retention.

The proposed SAKE Employee Benefits Program would provide eligible couriers with a personalized benefits card, offering a monthly allowance of UAH 350 to spend exclusively at SAKE Restaurant locations.

Each card would have a unique identification number and a scannable code linked to the employee's record in the company's database, eliminating the need to display personal information on the card.

The program aims to create additional value for couriers while keeping benefit-related spending within the restaurant network.

POTENTIAL VALUE

For couriers: Additional purchasing power and recognition.

For the business: A potentially cost-efficient employee benefit and an opportunity to support retention.

For operations: A supplementary measure addressing courier motivation alongside improvements to the delivery assignment process

ANASTASIIA PARUBETS

BENEFITS PROGRAM - Financial Feasibility Analysis

Internal Employee Benefits VS. Cash Compensation

Cash Compensation

Additional net compensation of UAH 350 per courier

554.55

UAH

Estimated total employer cost per courier/ month

Net amount for employee	350.00
Personal income tax %	81.82
Military tax %	22.73
Single social contribution 22%	100.00
Total employer cost	554.55

Sake Benefits Card

Monthly restaurant allowance worth UAH 350 per courier

234.50

UAH

Estimated total employer cost per courier/ month

Food production cost 45%	157.50
Card administration	20.00
Operational expenses	57.00
Total estimated cost	234.50

POTENTIAL ANNUAL COST DIFFERENCE

384,060

UAH

320.05

UAH saving per courier/ month

100

Couriers

1

Year

According to Article 165, Paragraph 165.1, Subparagraph 165.1.39 of the Tax Code of Ukraine (Law No. 2755-VI, dated December 2, 2010), non-cash gifts are exempt from personal income tax and military levy if their value does not exceed 25% of the statutory minimum monthly wage established on January 1 of the relevant tax year.

In 2026, this tax-exempt threshold amounts to UAH 2,161.75 per month.

Therefore, gift certificates valued at UAH 2,000 or less may qualify for exemption from personal income tax and military levy, provided that the relevant legal requirements are met.

CONCLUSION

This project began with a simple question: why was delivery performance deteriorating despite growing order volumes?

Through data preparation, SQL analysis and business process modelling, I identified courier assignment as a potential operational bottleneck.

The analysis revealed a significant increase in assignment times, accompanied by longer deliveries, higher cancellation rates and growing pressure on customer support.

Based on these findings, I developed an optimized courier assignment process designed to improve delivery efficiency while preserving fair opportunities for couriers.

I also proposed a controlled pilot to validate the solution and an additional employee benefits initiative to address potential courier motivation concerns.

Although the proposed solutions have not yet been implemented, the project demonstrates how data analysis and business process modelling can transform operational problems into structured, testable business recommendations.

The ultimate goal is not simply to identify what went wrong, but to understand why it happened and design a measurable path towards improvement (and save the money).

Potential Financial Value - UAH 384,060 / Year

THANK YOU

Any questions?

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